Mathematics, the video game

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Mathematics, a game?

According to the Oxford English Dictionary:[1]

II. An activity played for entertainment, according to rules, and related uses.

II.6.a. An activity or diversion of the nature of or having the form of a contest or competition, governed by rules of play, according to which victory or success may be achieved through skill, strength, or good luck. **Old English–**

Originally simply a specific use of sense I.2a, and sometimes not clearly distinguishable from it; in early use also chiefly restricted to indoor amusements, esp. those involving cards, dice, or playing-pieces, and to pastimes of a more or less light-hearted character. In later use sometimes distinguished from *sport* in not requiring physical exertion and skill, although in fact any rule-based competitive recreation may be referred to as a game, including activities more usually referred to as sports; cf. **sport** *n.*[1] I.4a.

In quot. OE as a partial gloss rendering Latin *ludus* 'sport', here denoting an athletic contest.

board game, *round game*, etc.: see the first element.

[1] https://www.oed.com/dictionary/game_n

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- Rules?

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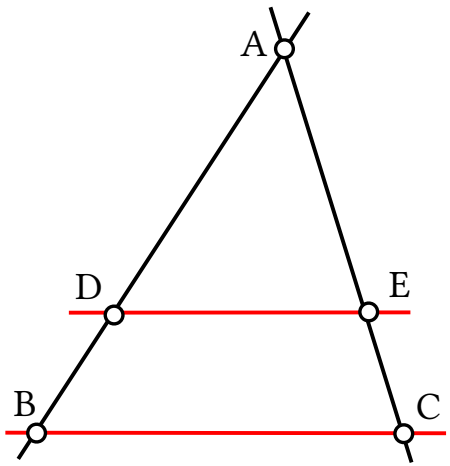

https://fr.wikipedia.org/wiki/Th%C3%A9or%C3%A8me_de_Thal%C3%A8s#/media/Fichier:Thales_theorem_1.svg

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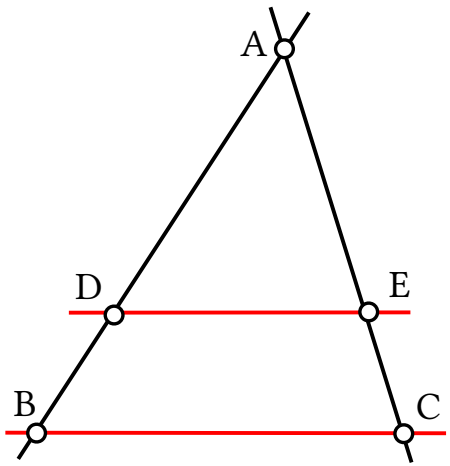

$$\int_2^{+\infty} \frac{1}{t \ln(t)} \, \mathrm{d}t = +\infty$$

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Mathematics, a game?

Two “sets of rules”: set theory and type theory

$$\forall x \forall y [\forall z (z \in x \leftrightarrow z \in y) \to x = y] \qquad \text{(Extensionality)}$$

$$\forall x \forall y \exists z \forall w (w \in z \leftrightarrow (w = x \vee w = y)) \qquad \text{(Pairing)}$$

$$\exists x [\exists y (\forall z (z \in y \to \mathbf{0}) \wedge y \in x) \wedge \forall w \in x \exists u \in x (w \in u)] \qquad \text{(Infinity)}$$

$$\forall x \exists y \forall z (z \in y \leftrightarrow \exists w \in x (z \in x)) \qquad \text{(Union)}$$

$$\forall x \exists y \forall z (z \in y \leftrightarrow \forall w \in z (w \in x)) \qquad \text{(Power Set)}$$

$$\forall p_0 \cdots \forall p_n \forall x \exists y \forall z (z \in y \leftrightarrow z \in x \wedge \varphi(z, p_0, \ldots, p_n)) \qquad \text{(Separation}_\varphi\text{)}$$

$$\forall p_0 \cdots \forall p_{n-1} \forall x [\forall y \in x \exists z \varphi(y, z, p_0, \ldots, p_{n-1}) \to \exists w \forall v \in x \exists u \in w\ \varphi(v, u, p_0, \ldots, p_{n-1})] \qquad \text{(Collection}_\varphi\text{)}$$

$$\forall p_0 \cdots \forall p_n \forall x [\forall y \in x\ \varphi(y, p_0, \ldots, p_n) \to \varphi(x, p_0, \ldots, p_n)] \to \forall z \varphi(z, p_0, \ldots, p_n) \qquad \text{(Set Induction}_\varphi\text{)}$$

Löwe, Benedikt & Paßmann, Robert & Tarafder, Sourav. (2021).

Constructing illoyal algebra-valued models of set theory. Algebra universalis.

82. DOI : 10.1007/s00012-021-00735-4.

Prod-Prop

$$\frac{E[\Gamma] \vdash T : s \qquad s \in \mathcal{S} \qquad E[\Gamma :: (x : T)] \vdash U : \mathsf{Prop}}{E[\Gamma] \vdash \forall x : T,\ U : \mathsf{Prop}}$$

Prod-Set

$$\frac{E[\Gamma] \vdash T : s \qquad s \in \{\mathsf{SProp}, \mathsf{Prop}, \mathsf{Set}\} \qquad E[\Gamma :: (x : T)] \vdash U : \mathsf{Set}}{E[\Gamma] \vdash \forall x : T,\ U : \mathsf{Set}}$$

Prod-Type

$$\frac{E[\Gamma] \vdash T : s \qquad s \in \{\mathsf{SProp}, \mathsf{Type}(i)\} \qquad E[\Gamma :: (x : T)] \vdash U : \mathsf{Type}(i)}{E[\Gamma] \vdash \forall x : T,\ U : \mathsf{Type}(i)}$$

Lam

$$\frac{E[\Gamma] \vdash \forall x : T,\ U : s \qquad E[\Gamma :: (x : T)] \vdash t : U}{E[\Gamma] \vdash \lambda x : T.\ t : \forall x : T,\ U}$$

App

$$\frac{E[\Gamma] \vdash t : \forall x : U,\ T \qquad E[\Gamma] \vdash u : U}{E[\Gamma] \vdash (t\ u) : T\{x/u\}}$$

Let

$$\frac{E[\Gamma] \vdash t : T \qquad E[\Gamma :: (x := t : T)] \vdash u : U}{E[\Gamma] \vdash \mathsf{let}\ x := t : T\ \mathsf{in}\ u : U\{x/t\}}$$

https://rocq-prover.org/doc/master/refman/language/cic.html#id4

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Don’t play all night…

Mathematics, a video game

Many interfaces (a.k.a. Interactive Theorem Provers)

- ⊢ Automath, de Bruijn, 1967
- ⊢ Mizar, Trubulec et al., 1973
- ⊢ Isabelle/HOL, Pawlson et al., 1986
- ⊢ Rocq[1], Coquand et al., 1984-1989

⋮

- ⊢ Lean, de Moura et al., 2013

[1] formerly known as Coq

Beyond the game

The scientific process relies on a peer-review system:

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What could go wrong?

Beyond the game

> "A technical argument by a trusted author, which is hard to check and looks similar to arguments known to be correct, is hardly ever checked in detail."[1] Vladimir Voevodsky, IAS, 2014

> "It soon became clear that the only real long-term solution to the problems that I encountered is to start using computers in the verification of mathematical reasoning." idem

[1] "https://www.math.ias.edu/~vladimir/Site3/Univalent_Foundations_files/2014_IAS.pdf"

Beyond the game

Some achievements:

- The 4-color theorem, 2008
- The odd-order theorem, 2012
- Kepler's conjecture, 2014
- Liquid Tensor Experiment, 2022
- $\mathrm{BB}(5) = 47,176,870$, 2024

Let's play!

Today, a Natural Number Game[1]:

[1] inspired from https://adam.math.hhu.de/#/g/leanprover-community/nng4

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- ⊢ A minimal setting to start playing with Rocq as fast as possible.
- ⊢ Many things left untold: do not hesitate to try things out!
- ⊢ Two languages to choose from:
  - ⇒ Vanilla Rocq
  - ⇒ SsrReflect

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